Research Impact and Open Science

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Outline

- Recap of Session 3 – Results to Discussion
- Foundations of “Research Impact”
- Discuss methods for planning/documenting for research impact
- Practical ways of ensuring visibility of research

Recap

- In session 3 of this module, we looked at how to use the 5-Step MOVE model to describe the discussion
- The MOVE model contains 5 sequential steps Background → Summarize or report → Comment → Highlight Limitations → Recommendations
- The method is applicable both for quantitative and qualitative research methods

Recap 5-Step Move Model (Social Science)

Original research question (Introduction): *How do smallholder farmers describe and justify their decisions about whether to adopt a new drought-tolerant maize variety?*

Move 1 – Background: *This study set out to understand how farmers describe and justify their decisions to adopt or reject a new drought-tolerant maize variety.*

Move 2 – Summarizing key results: *Three themes emerged from the interviews: trust in extension officers, concern about the upfront cost of seed, and social proof, that is, farmers were strongly influenced by whether trusted neighbours had already tried the variety.*

Move 3 – Commenting on key results: *Social proof appeared more influential in farmers' own accounts than the formal extension messaging which the programme had relied on. This contrasts with prior adoption studies that emphasize economic rationality as the primary driver. Cost concerns were also more prominent among smaller landholders, likely reflecting their thinner margin for risk if the new variety underperformed.*

Move 4 – Limitations: *These themes come from a purposively sampled group of adopters and non-adopters in one district, so the findings describe patterns of reasoning rather than how common each view is across the wider farming population.*

Move 5 – Recommendations: *Extension approaches should incorporate peer-to-peer demonstration alongside formal messaging, directly answering the original research question by showing that farmer adoption of new variety is substantially a social, not just economic, process.*

Research Impact

Research impact refers to the demonstrable contribution that research makes beyond academia to policy, practice, economy, environment, and society

Output→ Outcome→ Impact

Research Impact is distinct from academic output and outcome.

OUTPUT

What is produced

A paper on drought-resistant maize varieties

OUTCOME

Immediate change in knowledge, attitude, capacity

Farmers adopt the new variety after training

IMPACT

Longer-term, wider societal or environmental change

Measurable reduction in crop failure; improved food security

Domains of Impact

- **Contribution** → The areas of research impact including economy, environment, health, quality of life, community, organization etc.
- **Avenues** → Mechanisms or Processes by which research could have impact e.g. Effects on knowledge, understanding, awareness and/or attitudes
- **Change** → Effect or benefit to the area of influence
- **Levels** → Scale e.g. individual, local national, international

Alla, K., Hall, W. D., Whiteford, H. A., Head, B. W., & Meurk, C. S. (2017). How do we define the policy impact of public health research? A systematic review. *Health research policy and systems*, *15*(1), 84. https://link.springer.com/content/pdf/10.1186/s12961-017-0247-z.pdf

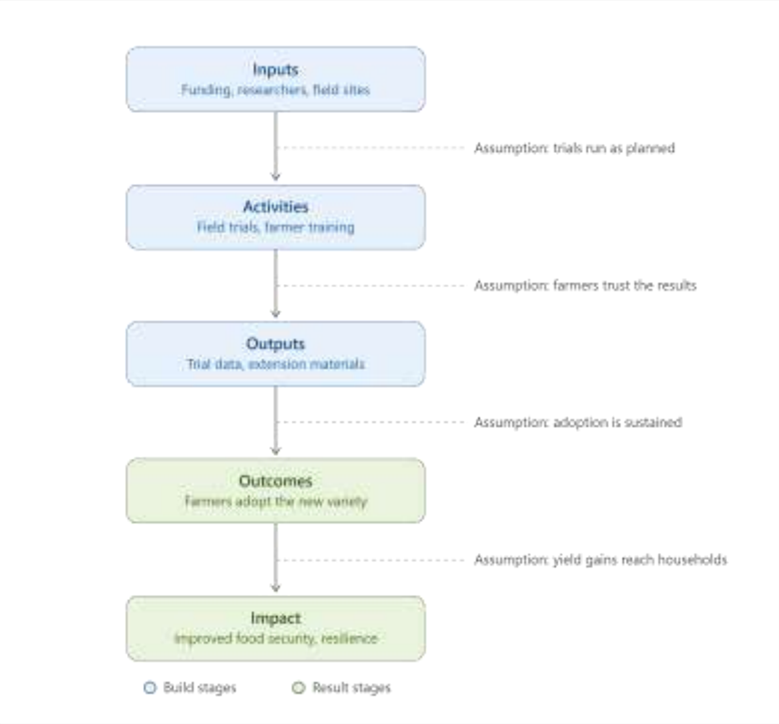
Frameworks for planning and tracing impact

- **Logical Framework Approach**: Matrix linking inputs, activities, outputs, outcomes, and goal — with indicators and assumptions
- **Theory of change**: Matrix linking inputs, activities, outputs, outcomes, and goal — with indicators and assumptions
- **Outcome mapping**: Focuses on behavior change in "boundary partners" rather than distant, hard-to-attribute impacts
- **Contribution analysis**: Builds a credible, evidence-based case for research's plausible contribution alongside other factors

Sample Log frame

Narrative Summary	Objectively Verifiable Indicators (OVIs)	Means of Verification (MOV)	Assumptions
GOAL (IMPACT)			
Improved food security and climate resilience for smallholder farming households	% reduction in household food insecurity in target districts within 3 years of variety release	District food security surveys; national agricultural statistics	Broader food system factors (markets, infrastructure) do not undermine household-level gains
PURPOSE (OUTCOME)			
Smallholder farmers adopt the drought-tolerant maize variety in place of the local check variety	% of farmers in target sites growing the new variety by end of Year 2; average yield loss under drought compared to local check	Household adoption surveys; follow-up yield sampling	Farmers who adopt continue planting the variety across multiple seasons, not just a single trial season
OUTPUTS			
1. Field trial evidence on yield performance under drought 2. Farmers trained on variety management 3. Extension materials disseminated	Number of trial sites completed; number of farmers trained (disaggregated by gender); number of extension materials distributed	Trial site reports; training attendance registers; distribution logs	Extension officers deliver training as designed; materials reach intended farmers
ACTIVITIES			
1.1 Conduct randomized field trials across 3 sites 2.1 Deliver farmer training sessions 3.1 Develop and print extension materials	Trial protocol adherence rate; number of training sessions delivered on schedule	Field diaries; session logs; budget/expenditure records	Timely funding disbursement; favorable weather allows trials to proceed as planned

Sample Theory of Change



- **Inputs → Activities → Outputs** are the "build" stages — the parts a research team directly controls (funding, fieldwork, training materials).
- **Outcomes → Impact** are the "result" stages — changes in the world that depend on other people acting (farmers adopting, households benefiting), which the project can influence but not fully control.
- Each arrow carries its own **assumption** — the thing that has to hold true for the pathway to keep moving forward.

Measuring impact, example

Domain	Example Indicator
Policy	Adaptation policies/bills citing the research
Practice	Farmers/communities adopting a practice from the research. Use of new tools from the research
Capacity	Researchers trained and using tools from research
Resilience	Reduction in crop loss or displacement based on adoption of the knowledge from research
Finance	Finance mobilized citing the evidence

Pitfalls undermining research impact

- Treating dissemination as equivalent to impact
- Designing impact plans only after research is complete
- Over-claiming attribution for multi-actor outcomes
- Producing policy briefs with no follow-up engagement
- Ignoring negative or null results
- Under-investing in translation capacity

Principles of maximizing Research Impact

- Design for impact at the onset
- Invest in translation capacity
- Build durable relationships with policy & community actors
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- Track outcomes, not just outputs
- Plan for resource mobilization as part of impact strategy

Assignment

1. Map the primary users, influencers, and beneficiaries of your research outputs

2. Sketch log frame of research question to include activities, outputs, out comes and impact. Also include the verifiable indicators, means of verification and assumptions

Questions and discussions